

Andre Arcaina

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Education

Toronto Metropolitan University (TMU, formerly Ryerson University)

Sept 2022 – Present

Bachelor of Science (Honours), Computer Science (Co-op)

Toronto, ON

Coursework: Data Structures, Algorithms, Database Systems, Computer Networks, Operating Systems, Software Engineering, Object Oriented Programming, Distributed Systems, Big Data Systems

Work Experience

Environment and Climate Change Canada (ECCC)

June 2025 – Present

Software Engineer Intern

North York, ON

- Built an internal Go-based end-to-end testing and observability tool for mission-critical satellite data pipelines, automating input and log tracing to significantly reduce manual bug triage effort.
- Modernized mission-critical Java services by migrating 8 components to Java 17 and Jakarta EE 10, improving maintainability and supporting national weather data delivery systems used in maritime and Coast Guard operations.
- Eliminated decades of technical debt by replacing a legacy VB6 application with a Python/PyQt dashboard, migrating the data layer to PostgreSQL to securely process datasets exceeding 50,000 rows.
- Reverse-engineered an undocumented legacy C++ and Java system, authoring foundational architecture and UML diagrams to accelerate future feature development for a 3-person team.

Undergraduate Science Society of TMU (USSTM)

Jan 2025 – Present

Backend Engineer

Toronto, ON

- Engineered a contract-first backend API in Go powering platform services for the undergraduate science student body, leveraging SQLc for type safety and OpenAPI to enforce strict definitions across 32+ endpoints.
- Developed 37+ unit/integration tests achieving 85%+ code coverage, preventing regressions in core business logic.
- Built AWS S3 image storage with presigned URLs to support secure, horizontally scalable before and after item image uploads.
- Developed asynchronous email processing using Redis-backed Async workers and AWS SES for reliable notification delivery.

DataKinetics

May 2024 – Aug 2024

Software Engineer Intern

Ottawa, ON

- Developed a Spring Boot middleware service and a web dashboard, utilizing an external Java library to automate COBOL-to-JSON parsing and eliminate manual data inspection for the R&D team.
- Presented the finalized tool to 12 R&D stakeholders, successfully demonstrating the new automated data-inspection workflow.
- Designed and executed 72 comprehensive Postman test cases for a core mainframe REST API, rigorously validating JSON responses, edge cases, and data integrity across 3 distinct IBM Db2 database tables.

Projects

Fafnir, Distributed Paper Trading Platform | Go, gRPC, Docker, Kubernetes, NATS, PostgreSQL, Redis

GitHub

- Built a distributed backend for a fintech paper trading platform composed of 8+ microservices, handling 1,600+ RPS across 5,000 concurrent users during load testing with a 0% failure rate (P95: 42ms, Avg: 7ms).
- Deployed infrastructure using Kubernetes and Helm, integrating system observability with Prometheus, Loki and Grafana.
- Improved market data processing latency by 98% (800ms to 10ms) using Redis caching and request coalescing.

Hyperion, Distributed Key-Value Database | Go, Raft, gRPC, BadgerDB

GitHub

- Engineered a highly available key-value store in Go, utilizing the Raft algorithm to guarantee zero data loss and reliable backups across multiple node processes.

Specter, Autonomous Legal Agent | Python, FastAPI, ChromaDB, Voyage AI, OpenClaw (Claude)

GitHub

- Built an agentic RAG system indexing 724 statutes across 7 jurisdictions, enabling lawyers to query case scenarios via WhatsApp and receive grounded insights in seconds, reducing research time.

Skills

Languages: Go, Python, Java, C++, Rust, JavaScript, TypeScript, SQL, GraphQL

Frameworks: FastAPI, Django, Spring Boot, Node.js, React.js, Next.js, TailwindCSS, gRPC

Tools: GCP, AWS, Docker, Kubernetes (Minikube), NATS, Postgres, Redis, BadgerDB, Git, Bash, Jenkins, GitHub Actions